Speed Adaptive Probabilistic Flooding in VANETs

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Abstract

Vehicular Ad Hoc Networks are emerging as the preferred network design for intelligent transportation systems and are envisioned to be useful in road safety and commercial applications. A significant issue in VANETs is the design of an effective broadcast scheme which can facilitate the fast and reliable dissemination of critical safety messages to neighbouring vehicles in case of an unexpected event, such as a traffic accident. Towards to this goal, we propose a novel Speed Adaptive Probabilistic Flooding algorithm (SAPF).[1]Upon receiving an emergency message, a vehicle decides to rebroadcast the message with propability p whose value is chosen based on the speed of the vehicle. The algorithm enjoys a number of benefits relative to other approaches: it is simple to implement and does not introduce additional communication burden as it requires local information only, it does not rely on the existence of a positioning system which may not always be available and above all, mitigates the effect of the broadcast storm problem, typical when utilizing blind flooding. We design the algorithm and evaluate its perfornance using an integrated platform combining the OPNET Modeler and VISSIM simulators. Our results indicate that the proposed algorithm outperforms blind flooding, especially in cases of heavy congestion. The SAPF algorithm achieves high reachability and unlike blind floding, maintains low latency as the density of vehicles in the road network increases.

I. Introduction

Thousands of injuries and hundreds of fatalities are reported daily worldwide due to traffic accidents [1]. In an attempt to reduce these terrifying numbers, government agencies and the private sector are responding by introducing Intelligent Transportation Technologies to the existing transportation system. Vehicular Ad hoc Networks are emerging as the preferred network design for ITS technologies. Vehicles employ wireless communication to form ad hoc networks which are envisioned to accommodate the new generation of cooperative road safety applications. A popular example of such an application is cooperative emergency warning. Upon detection of an unexpected event such as a traffic accident, a vehicle generates an emergency warning message which it disseminates to neighboring vehicles through the underlying network. Wireless multi-hop communication extends the line of sight of the drivers and promises to reduce the delay in propagating the emergency warning message. A major challenge in such an application is the design of an efficient broadcast scheme which will facilitate the fast and reliable dissemination of the early warning message to the approaching vehicles.

A straightforward solution is flooding [2], an approach which involves each vehicle rebroadcasting the message whenever it receives it for the first time. However, blind flooding is known to generate a large number of redundant messages leading to contention and unnecessary collisions. This is known as the Broadcast Storm Problem [2]. Several techniques have been proposed in literature to alleviate this problem [3-12]. The main idea has been to reduce the number of nodes rebroadcasting the message without affecting the total number of nodes receiving the message. The various proposed solutions then differ in the method with which this restricted set of nodes is chosen.

Specifically for VANETs, the most popular approach has been to choose vehicles which lie on the boundary of the transmission range of the vehicle transmitting each message [13-15]. However, this method assumes the availability of a positioning system, such as GPS, which may not always be available. Our main contribution in this work is to develop a new broadcast scheme which does not rely on the existence of a positioning system and is shown to work effectively in a number of scenarios outperforming other approaches. In order to reduce the number of vehicles rebroadcasting the message we employ probabilistic flooding and in addition, we adaptively regulate the rebroadcast probabilities, based on the vehicle speeds, to optimally reduce message delivery delays caused by increased contention, in areas with high vehicle densitites. We refer to the proposed scheme as Speed Adaptive Probabilistic Flooding (SAPF). The main idea is that each vehicle upon receiving an emergency message for the first time, decides to rebroadcast the message with propability p whose value is calculated as a function of the speed of the vehicle. This function aims at mapping the speed of the vehicle to the optimal rebroadcast probability value which coincides with what is known as the ´critical´ probability. As we show in this paper probabilistic flooding in VANETs is characterized by phase transition phenomena which dictate optimal values for the rebroadcast probability at the onset of the transitions. At these optimal values high reachability is achieved with the minimum possible latency. These optimal values, however, depend on the vehicle densities which in turn are known to depend on the vehicle speeds [22]. In this paper we investigate these phase transition phenomena and the foremntioned relationships and we derive a piecewise linear function relating the rebroadcast probability with the vehicle speed. This function constitutes the basis of the proposed SAPF scheme.

The algorithm enjoys a number of benefits relative to other approaches: it is simple to implement and does not introduce additional communication burden as it requires local information only, it does not rely on the existence of a positioning system which may not always be available and above all, mitigates the effect of the broadcast storm problem, typical when utilizing blind flooding. The scheme is evaluated using an integrated platform combining the OPNET Modeler and VISSIM simulators. We have chosen a realistic reference model to conduct our study based on a section of the Limassol-Nicosia

[1] This work was partly funded by RPF TRAFFICNET.

freeway in Cyprus. The traffic model was generated on the VISSIM simulator and was used to produce traces of the vehicles involved. These traces where then fed into the OPNET Modeler simulator, which was used to model the networking aspects of the proposed system. Our results indicate that the proposed algorithm outperforms blind flooding, especially in cases of heavy congestion. The SAPF algorithm achieves high reachability and unlike blind floding, maintains low latency as the density of vehicles in the road network increases.

The paper is organized as follows. In section II, we give a brief introduction to phase transition phenomena, percolation theory and random graph theory to show the correlation between these theories and VANETs. In section III, we present the rationale behind our design choices and the adopted design methodology and in section IV we present the system architecture and implementation details of the proposed SAPF algorithm. In Section V we evaluate the performance of the proposed scheme using simulations and finally in section VI offer our conclusions and future research directions.

II. Phase Transition, Percolation Theory and Random Graphs

P*hase transition* or *phase change* is a phenomenon where a system undergoes a sudden change of state or behavior in response to a small change in a chosen parameter set [19]. The phenomenon is of great importance as it has been observed and analyzed in a number of systems pervading our physical and engineering world. Two areas of research where phase transition has been extensively studied are percolation theory and random graphs. The study in these areas has led to abstract mathematical frameworks which can serve as solid starting points to investigate other systems where phase transition is relevant, such as probabilistic flooding in vehicular ad hoc networks.

Percolation theory deals with fluid flow (or any other similar process) in random media. There are two types of percolation models, *site percolation* and *bond percolation,* see Fig 1. A *site percolation* considers the lattice vertices as the relevant entities. A *bond percolation* considers the lattice edges as the relevant entities.

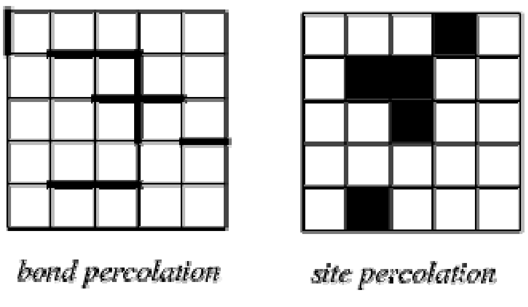

Figure 1: Bond & Site Percolation

In both models there is a probability *p* that there is an open path and 1-p that there is no path. It has been observed that phase transition can occur in percolation models at the change of state between having finite numbers of clusters and having one infinite cluster. The existence of a phase transition in percolation theory infers the existence of a critical threshold probability pc value where the phase transition occurs. It will be interesting to investigate the possible application of the same theory to the probabilistic flooding problem in VANETs. Such an application can provide valuable insights to the observed behavior and can also lead to analytically verifiable designs.

Phase transition has also been observed and studied in the context of Random Graphs. In mathematics, a random graph is a graph that is generated by some random process [20]. Different choices of the random process yield different random graph models. Various models have been studied in literature such as the Fixed Edge Number Model, the Bernoulli model, the Fixed Radius model and the Dynamic model. In the case of a Bernoulli model with N vertices and probability of an edge existence between any two nodes equal to p, it has been shown that for values of p greater than log(N)/N, the graph is connected with high probability [19]. This result not only verifies the existence of phase transition phenomena in random graphs but also quantifies the critical point. The problem of obtaining similar results for the Fixed Radius model which is an ideal representation of Mobile Ad Hoc Networks is a challenging open research problem.

Since random graphs are relevant to wireless ad hoc networks, it is not surprising that recently, phase transition phenomena have been reported in various contexts in wireless ad hoc networks. In the context of probabilistic flooding, phase transition can be extremely cost-efficient to observe, as it implies that there exists a critical probability beyond which all nodes receive the transmitted message with high probability. Consider the scenario where a node in a wireless ad hoc network broadcasts a query message and then each node rebroadcasts the message with probability p. In Fig. 2 [21] we show a graph of the probability that the message reaches all nodes versus the probability of rebroadcasting the message. We observe that there exists a certain threshold, which we refer to as the critical threshold, beyond which all nodes receive the message with high probability. For probability values above 0.55, all vehicles receive the message with probability greater than 0.9. It has been pointed out [21] that this critical threshold depends on the node density. Our objective in this work has been to verify that such phase transition phenomena are also observed in vehicular ad hoc networks, demonstrate how traffic density affects the critical threshold values and examine how our findings can help in the design of an effective broadcast scheme employing probabilistic flooding.

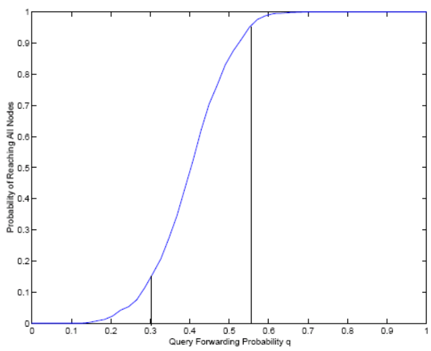

Figure 2: Phase Transition Probability

III. Design Rationale

In this section we present the rationale behind our design choices and the adopted design methodology. Our objective has been to design an effective broadcast scheme which facilitates reliable and fast delivery of emergency messages to neighboring vehicles in case of an unexpected event such as a traffic accident. Probabilistic flooding was chosen as the candidate solution in order to mitigate the effects of the broadcast storm problem encountered when using blind flooding. In order, however, to achieve optimal performance, we have also decided to set the rebroadcast probability at each vehicle adaptively, based on the vehicle speed. The reasoning behind this choice is the following: low vehicle speeds in a freeway setting imply high vehicle density, which then imply that high reachability can be achieved by choosing relatively low rebroadcast probability values. Probabilistic flooding in mobile ad hoc networks is known to yield phase transition phenomena. As the rebroadcast probability increases, there exists a threshold value beyond which reachability suddenly reaches 100 percent with high probability.

This critical value is the rebroadcast probability of choice as it guarantees high reachability and low delay. If higher rebroadcast probabilities are chosen high reachability is still achieved, however, delays may increase due to more vehicles rebroadcasting the emergency messages. However, the critical probability depends on the node density. In a vehicular setting, the vehicle density is also known to relate to the vehicle speed. So, by suitable choice of the rebroadcast probability as a function of the vehicle speed optimal performance can be achieved at all traffic densities. In the remainder of this section, we show the method adopted to derive the desired relationship between the rebroadcast probability and the vehicle speed. We first use simulations to find the critical probability at various vehicle densities and we then use known relationships between the vehicle density and the speed to derive the desired relationship between the rebroadcast probability and the vehicle speed.

All the simulation experiments were conducted using an integrated platform combining two simulators, VISSIM, a traffic simulator, and OPNET Modeler, a network simulator. The assumed setting is the following: A vehicle that detects an unexpected road hazard becomes an abnormal vehicle (AV) and transmits instantly an early warning message (EWM) to warn approaching vehicles of the imminent danger. Following the specifications of the 802.11b standard, the transmission range of all vehicles is set to 300 meters. Upon reception of the EWM for the first time a vehicles decides to rebroadcast the message with probability p, or decides not rebroadcast the message with probability $1-p$.

The probabilistic flooding algorithm is as follows:

Probabilistic Algorithm

1. **upon** reception of EWM ***s*** at node ***n***:
2. **If** EWM received first time and it is not the originator of message ***s*** and TTL > 0 then
3. broadcast EWM ***s*** with probability ***p***
4. **end if**

The performance metrics that we consider in our study are the number of nodes which receive the transmitted emergency message, the protocol overhead, and the latency. The latency is defined as the time interval between the instant the message is transmitted by the vehicle detecting the road hazard and the instant that the last vehicle in the network receives the message. We use two measures for the protocol overhead, the number of backoffs and the number of rebroadcasts. A node enters a backoff state before each transmission and after the medium is sensed to be non-idle and is an indicative measure of the protocol contention.

The chosen test site is a two lane highway which spans a distance of 6Km. We conducted a number of experiments to reflect different scenarios. Our objective has been to examine the behavior of the above performance metrics as we change the rebroadcast probability and the vehicle density. We have considered rebroadcast probability values in the range 0 to 1 in steps of 0.1. To generate scenarios with different vehicle densities we considered different rate of vehicles entering the chosen test site. We considered the following penetration rates: 5 vehicles per kilometer per lane, 10 per kilometer per lane, 20 per kilometer per lane, 30 per kilometer per lane, 40 per kilometer per lane and 50 per kilometer per lane. All values obtained are averages over 10 simulation experiments.

Fig. 3 shows the percentage of vehicles receiving the emergency warning message as we change the rebroadcast probability and the vehicle penetration rate. For a particular rate it is evident that there exists a critical threshold probability beyond which the number of vehicles receiving the message suddenly increases and stays almost constant. This sudden change is compatible with phase transition phenomena observed in the literature of random graphs and percolation theory. We also observe that as the penetration rate and thus the vehicle density increases, the critical threshold decreases. It is apparent that almost all vehicles receive the EWM for probabilities above 0.5. Consequently, in order to make a more detailed investigation of the behavior of probabilistic flooding in probability values below 0.5, we repeated the simulation experiments with probability values in the range 0 to 0.5 with steps of 0.05. The results are presented in Fig 4 which allows one to obtain more accurate critical probability values. The obtained critical probability values are plotted in Fig. 5 vs the corresponding penetration rates. We observe an almost linear decrease of the critical probability as the penetration rate increases up to a saturation point (50 per kilometer per lane) after which the critical probability stays constant. This saturation is due to the fact that after some value of the penetration rate the vehicle density remains almost constant since the capacity of the particular road has been reached. For a particular section of the road there is a maximum number of vehicles which can be present simultaneously in that section.

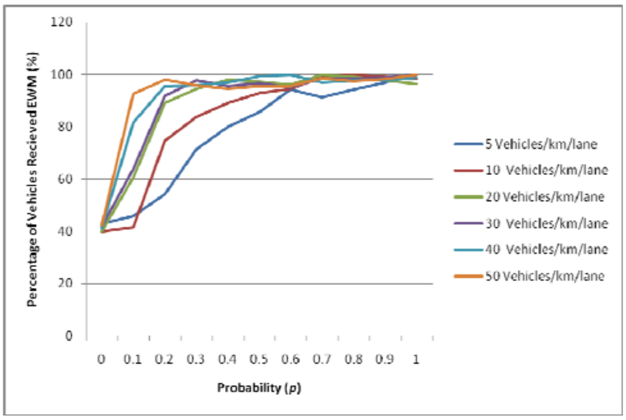

Figure 3: Percentage of Nodes Received EWM vs. Probability

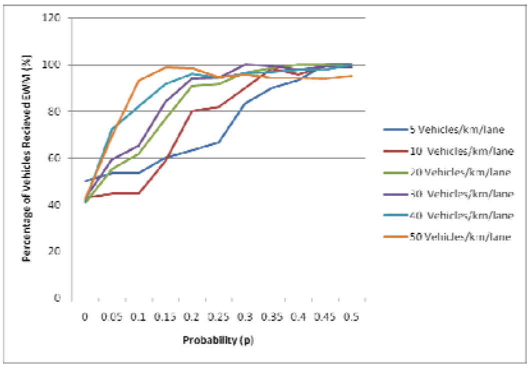

Figure 4: Percentage of Nodes Received EWM vs. Probability

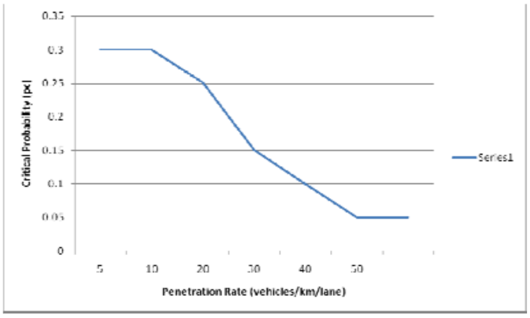

Figure 5: Critical Probability vs. Vehicle Densities

Our next objective is to find the desired rebroadcast probability value at each vehicle density. The desired value is the one which achieves high reachability but minimizes the overhead and thus the delays. We use two measures for the overhead. The number of backoffs and the number of message rebroadcasts. In Fig. 6, we present the number of backoffs reported as we vary the rebroadcast probability and the penetration rate of the vehicles. We observe similar behavior at all rates with the number of backoffs increasing as we increase the rebroadcast probability. This is expected as higher rebroadcast probabilities imply a larger number of vehicles contenting for the common medium.

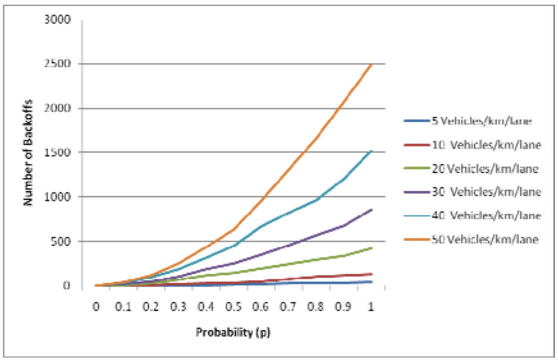

Figure 6: Number of Nodes Backoffs vs. Probability

The greater contention observed as we increase the penetration rate is caused primarily due to the larger number of vehicles trying to rebroadcast the message. This is verified in Fig. 7 where we plot the number of rebroadcasts versus the penetration rate. We observe that at all penetration rates as we increase the rebroadcast probability the number of vehicles rebroadcasting the message increases as well.

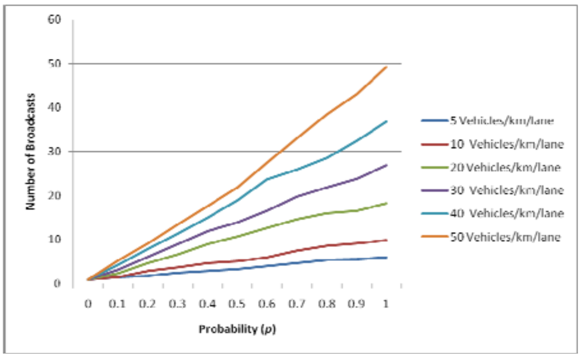

Figure 7: Number of Nodes Broadcasts vs. Probability

Finally in Fig. 8 we show plots of the message propagation delay. The graphs indicate that at all penetration rates the delay exhibits a monotonically increasing pattern. This is also expected as higher contention causes collisions which delay message delivery.

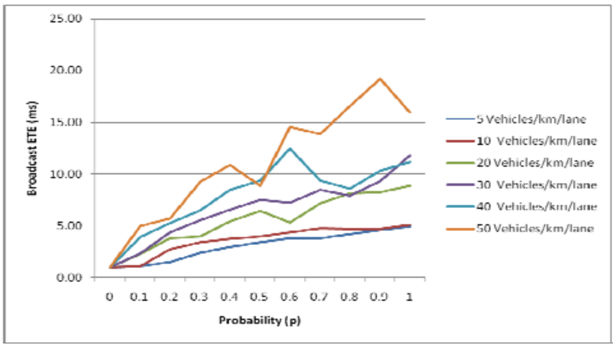

Figure 8: Broadcast ETE vs. Probability

What the above graphs indicate is that as we increase the rebroadcast probability beyond the critical probability, the reachability remains high while the latency increases. So, for a particular node density the critical probability is the optimal choice for the rebroadcast probability which ensures the fastest message delivery to almost all vehicles. The next step is to find the relationship which allows each vehicle to calculate this optimal value based on its own speed. We derive the relationship by composition of the mapping shown in Fig. 5 with the vehicle speed vs. density relationship shown in Fig. 9 [22]. The latter was obtained from field data. The resulting function, mapping the speed to the rebroadcast probability is shown in Fig. 10. We consider a linear approximation of the graph also shown in Fig. 10 whose equation is given by:

$$p = 0.0557x - 0.003 \qquad (1)$$

where p denotes the rebroadcast probability and x the velocity of the vehicles. In addition, for speeds above 100km/h we set the rebroadcast probability to 1. The reason is that beyond such speeds it is impossible to estimate the density based on speed information only and so we adopt a rather aggressive rebroadcast policy in order to ensure message delivery to all vehicles. On the other hand for speed values below 15km/h, we assume that the network has almost reached its capacity and so we consider a constant rebroadcast probability of 0.05. The resulting rebroadcast probability vs. speed function used in the proposed SAPF scheme is shown in Fig. 11.

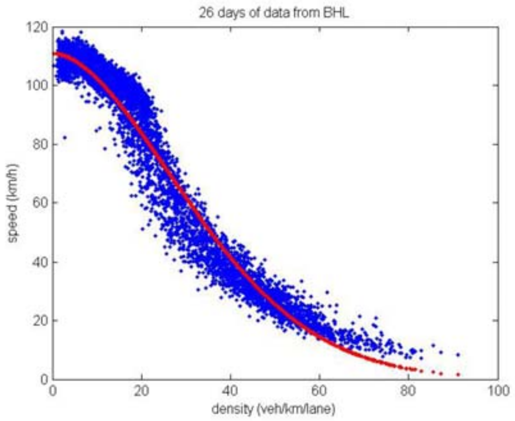

Figure 9: Vehicle speed vs. Density

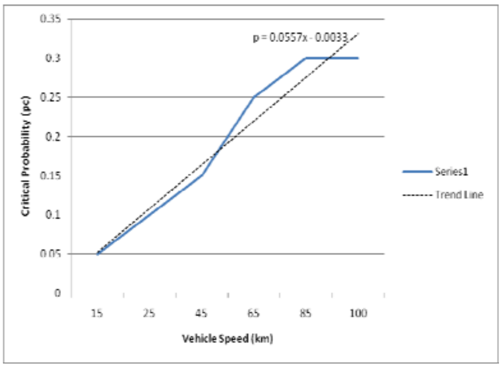

Figure 10: Critical Probability vs. Vehicle speed

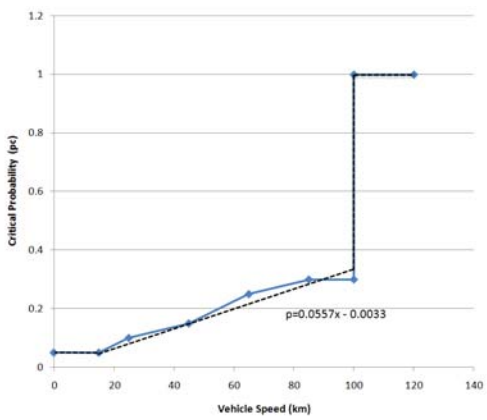

Figure 11: Critical Probability vs. Vehicle speed

IV. System Overview and Implementation

In this section, we give a brief overview of the assumed VANETs technology and we present implementation details of the SAFP algorithm,. The PHY/MAC layers of VANETs are based on the 802.11 standards. In USA, there is currently an effort to standardize the 802.11p for vehicular ad hoc networks. IEEE 802.11p is based on orthogonal frequency-division multiplexing (OFDM) but it uses 10MHz channels instead of 20MHz. channels as in 802.11a. The IEEE 802.11p operates in 5.8/5.9GHz band and use a simple CSMA scheme. The data rates that can be obtained per channel are from 3 to 27 Mb/s.

The setting that we consider is similar to the one presented in section III. At the instance of the reception of an EWM for the first time, vehicles decide to rebroadcast the message with probability *p*, or not rebroadcast the message with probability 1-*p*.

The AV (vehicle that detects an unexpected road hazard) is the originator of the EWM message and remains the originator of the message until the message expires or is destroyed. Every message has the following fields: IP address of the originator (AV), destination address (set to broadcast), a time to live field (TTL), and sequence number field. All these fields are set by the AV at the time of the creation of the EWM message. The only field that can be modified by other vehicles is the time to live field, TTL. The TTL field is decreased by one every time the EWM is rebroadcasted. As soon the TTL reaches a zero value the message is destroyed. The TTL depicts the number of hops the EWM can be transmitted in the network. Each vehicle maintains a table of recently received messages. Each message is identified by a combination of the source IP address and the sequence number. Whenever a vehicle receives a new message, it checks whether this message has been recently received through matching of the identifiers in the relevant table. If no matching is found, the message is classified as being received for the first time and is made available for rebroadcast upon decision of the probabilistic algorithm. The identifier of the message is also added to the table of recently received messages. This table may include up to 99 entries.

Based on the function of Fig. 11 derived in the previous section, the speed adaptive probabilistic flooding algorithm is as follows:

Speed Adaptive Probabilistic Flooding Algorithm

1. upon reception of EWM *s* at node *n*:
2. if EWM received first time and it is not the originator of message *s* and TTL > 0 then
3. if (veh. speed >= 15km && veh. speed <= 100km) then
4. broadcast EWM ***s*** with probability **p** = 0.0557x-0.0033
5. else if (veh. speed < 15km) then
6. broadcast EWM ***s*** with probability **p** = 0.05
7. else if (veh. speed > 100km) then
8. broadcast EWM ***s*** with probability **p** = 1
9. end if
10. end if
11. end if
12. end if

For the implementation of the SAFP algorithm we used OPNET Modeler simulation engine by making appropriate modifications to an 802.11 WLAN station model which excludes implementations of layers above the MAC layers such as TCP/IP. The model is implemented in the Wlan_station_adv mobile node module which emulates layers above the MAC by appropriate source and sinks modules. The most significant changes were made to the *wireless_lan_mac* module where the MAC layer is implemented. The Mac layer differentiates the packets that receives from the physical layer and the packets that receives from application layer. A time to live field has been added to the header of the transmitted packets. Upon receiving a

packet from the application layer, the MAC layer sets the TTL value equal to a predefined TTL_max value. Whenever a packet is received from the physical layer, the TTL field is decreased by one and it transmitted with probability p. In addition, the IP address of the source of the packet is maintained in the packet header and is not updated with the IP address of the node receiving the message as was the original implementation. This is done in order to create appropriate identifiers for each packet which will allow each node to infer whether it has received the warning message for the first time. These unique identifiers are created by combining the IP address of the packet source with the source sequence number which is also included in the packet header. A table of recently received messages has also been created in the module. When receiving a message, a node checks whether the identifier of the message is included in the table. If it is included, then the packet is destroyed. If it is not, then a new entry is added in the table and the packet is made available for retransmission. The packet is retransmitted with probability *p*. The probability p is calculated based on the SAFP algorithm.
The TTL_max and transmission probability parameters could be set separately for each vehicle. Table 1 demonstrates the WLAN parameters chosen for all nodes.

WLAN Parameters	**Values**
Transmission Power	0.00043
Transmission Probability	Promoted
Time to Live	2
Data Rate	11 Mbits/s
Destination address	Broadcast

Table 1: Simulation Parameters

V. Performance Evaluation

In this section, we evaluate the performance of the SAPF algorithm using simulations. We conduct our study using an integrated platform combining two simulators, the traffic simulator VISSIM and OPNET Modeler. We have used VISSIM to generate traces of the vehicles involved in our scenarios. These traces were then fed into the OPNET Modeler simulation model which was used to simulate the networking aspects of the proposed solution. In Fig. 12 and 13 we show snapshots of the simulation models generated on the two simulators.

The topology of the simulation model used in our study is based on a section of the Nicosia-Limassol highway in Cyprus. The chosen test site is a two lane highway which spans a distance of 6Km. Our objective has been to evaluate the performance of the SAPF algorithm with r espect to chosen performance metrics and compare it with the Blind Flooding (BF) algorithm. The performance metrics considered, were the percentage of the number of nodes receiving the transmitted emergency message, the protocol overhead, and the latency. We conducted a number of experiments, in order to evaluate the performance of the SAPF scheme in scenarios reflecting three types of congestion: light congestion, regular congestion and heavy congestion. We emulate the different congestion levels in the chosen topology as follows: The vehicles in the considered two lane freeway propagate abiding to an original speed limit in the range 90-120km/h. In order to emulate an unexpected event which generates congestion, at the 3rd kilometer of the freeway we suddenly change the speed limit to lie in the range 30-60km/h. We then create different levels of congestion by varying the penetration rate. We considered three penetration rate values, 2000veh/hour, 5000veh/hour and 50000veh/hour reflecting light, regular and heavy congestion conditions respectively.

The network is confined within a 10m *x* 6000m area, where each vehicle has a constant transmission range of 300m. The latter is ensured by setting the Transmission Power equal to 0.00043w. Each vehicle uses 802.11b MAC layer protocol and operates in ad hoc mode. The vehicles are configured to operate in Broadcast mode with no ACK/CTS/RTS mechanisms. The TTL value is set to 2 in order to guarantee a maximum of 3-hop transmission. This creates a multi-hop transmission range of approximately 900m. The data size of each packet is set to a constant value of 1024 bytes. In all simulations, the early warning message is generated 400 seconds after the simulation start time. All simulations run for 1 hour network simulated time.

Figure 12: Snapshot of the reference model generated on the VISSIM Simulator

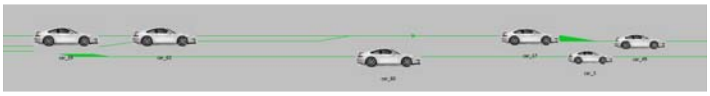

Figure 13: Snapshot of the reference model generated on the OPNET Modeler Simulator

Results

In this section we present the results obtained and we extract useful conclusions regarding the performance of the SAPF algorithm comparing it with the BF algorithm under light, regular and heavy congested traffic. Our results indicate that the SAPF algorithm outperforms the BF algorithms as it maintains high reachability and low delays at all congestion levels.

Fig. 14 shows the percentage of vehicles receiving the emergency warning message for the three penetration rates under consideration. It is evident, that both algorithms in all cases perform equally well achieving reachability values higher than 92%. The SAPF algorithm in fact achieves reachability values greater than 97% which is more than satisfactory.

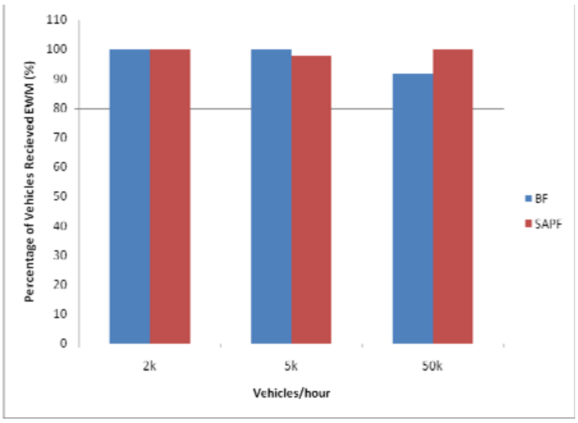

Figure 14: Percentage of Nodes Received EWM vs. Probability

In Fig. 15 we present the number of backoffs generated by the two algorithms under the three aforementioned scenarios.

We observe that at light congestion conditions the two algorithms report similar values. At more severe congestion conditions, however, the SAPF algorithm significantly outperforms the BF algorithm as it consistently reports smaller number of backoffs. As congestion becomes more severe, blind flooding generates increasing number of backoffs while the SAPF scheme maintains approximately the same numbers.

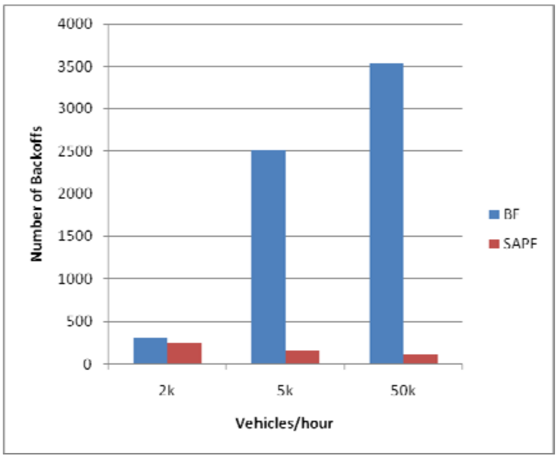

Figure 15: Number of Nodes Backoffs vs. Probability

The greater contention observed as we increase the traffic density is caused primarily due to the larger number of vehicles trying to rebroadcast the message. This is demonstrated in Fig. 16 where the number of rebroadcasts reported is shown for the three considered scenarios. The picture is almost identical to the one in Fig. 15.

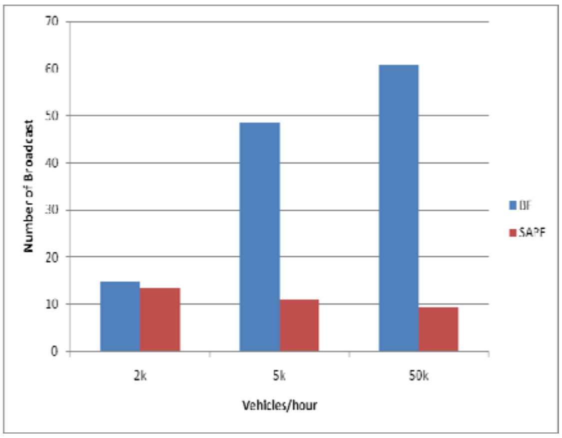

Figure 16: Number of Nodes Broadcasts vs. Probability

Finally in Fig. 17 we show plots of the message propagation delay. This graph highlights the true superiority of the SAPF scheme compared to the BF algorithm. As the congestion becomes heavier, the BF algorithm takes more time to deliver the messages to all vehicles. The SAPF algorithm, on the other hand, manages to maintain almost constant latency values at all congestion conditions. In the heavy congestion case for example we observe that blind flooding reports a delay of approximately 22ms while the SAPF reports a delay of approximately 7ms. This significant reduction in the delay is of great importance in the considered applications as it gives the drivers or the automatic controller additional time to respond effectively.

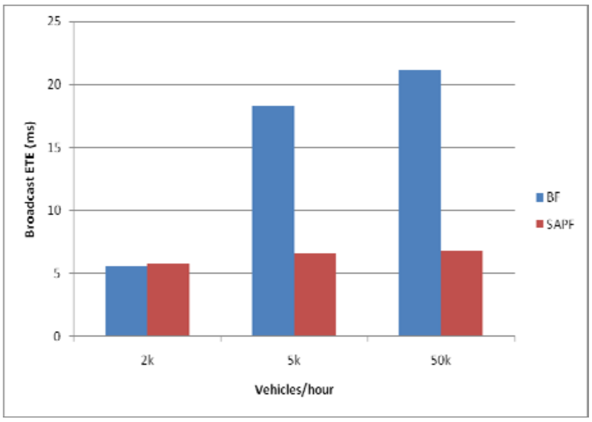

Figure 17: Broadcast ETE vs. Probability

VI. Conclusions

In this paper we present a novel broadcast scheme for VANETs which does not rely on the existence of a positioning system and is shown through simulations to work effectively in a number of scenarios outperforming other approaches. The scheme employs probabilistic flooding to reduce the number of vehicles rebroadcasting emergency messages and in addition it adaptively regulates the rebroadcast probabilities, based on the vehicle speeds, to optimally reduce message delivery delays caused by increased contention, in areas with high vehicle densitites. We refer to the proposed scheme as Speed Adaptive Probabilistic Flooding (SAPF). We evaluate its perfornance using an integrated platform combining the OPNET Modeler and VISSIM simulators. Our results indicate that the proposed algorithm outperforms blind flooding, especially in cases of heavy congestion. The SAPF algorithm achieves high reachability and unlike blind floding, maintains low latency as the density of vehicles in the road network increases. We aim at further evaluating the performance of the proposed scheme in more complex scenarios and in addition, verify our simulation results with analytical resutls which will be obtained using tools from random graph theory.

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